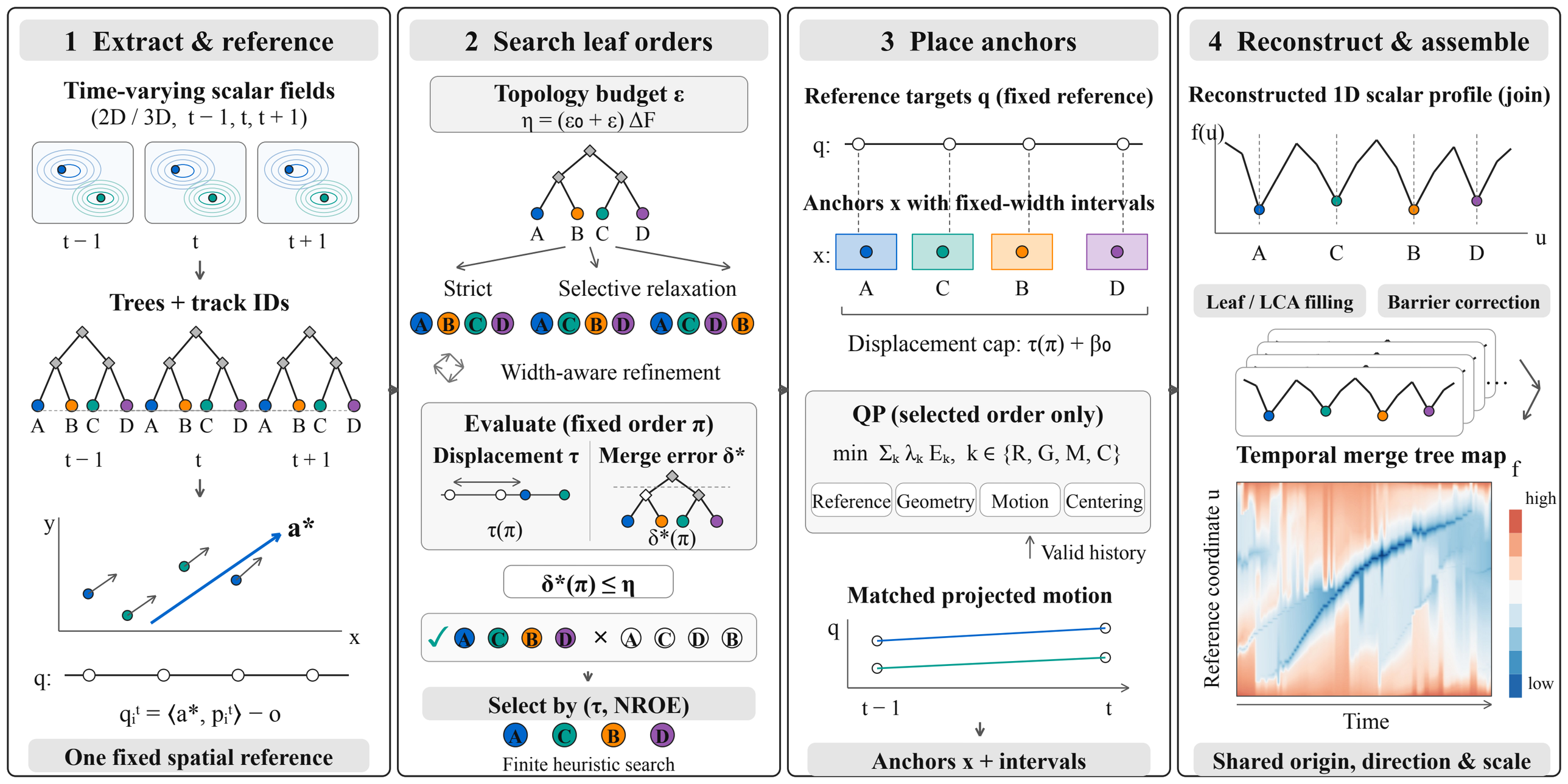

1 Extract & reference
Time-varying scalar fields
(2D / 3D, t − 1, t, t + 1)
t − 1
t
t + 1
Trees + track IDs
A B C D A B C D A B C D
t − 1
t
t + 1
y
a*
x
q:
$q_i^t = \langle a^*, p_i^t \rangle - o$
One fixed spatial reference
2 Search leaf orders
Topology budget ε
η = (ε0 + ε) ΔF
A B C D
Strict
Selective relaxation
A B C D
A C B D
A C D B
Width-aware refinement
Evaluate (fixed order π)
Displacement τ
Merge error δ*
τ(π)
δ*(π)
δ*(π) ≤ η
A C B D
A C D B
Select by (τ, NROE)
A C B D
Finite heuristic search
3 Place anchors
Reference targets q (fixed reference)
q:
Anchors x with fixed-width intervals
x:
A C B D
Displacement cap: τ(π) + β0
QP (selected order only)
min Σk λk Ek, k ∈ {R, G, M, C}
Reference
Geometry
Motion
Centering
Valid history
Matched projected motion
q
t − 1
t
Anchors x + intervals
4 Reconstruct & assemble
Reconstructed 1D scalar profile (join)
f(u)
u
A C B D
Leaf / LCA filling
Barrier correction
Temporal merge tree map
f
high
low
Reference coordinate u
Time
Shared origin, direction & scale